CBOT SOYBEAN FUTURES (ZS)

Spread & Curve Structure — Fundamental Reference

The Crush, China, Brazil, and the US-Brazil Seasonal Supply Handoff

This document presents a fact-based reference on the fundamental drivers of CBOT Soybean futures (ticker: ZS), contract size 5,000 bushels, with particular focus on the forces that shape the forward curve and the spread structure between contract months. Soybeans are distinguished from corn and wheat (covered in separate reference documents in this series) by several structural features: the crop is processed ("crushed") into two co-products — soybean meal and soybean oil — whose combined value determines processing economics in a three-way price relationship unique among the major row crops covered in this series; China is, by a wide margin, the single largest national demand centre for soybeans globally, making Chinese import policy and demand a more concentrated single-country variable than for almost any other commodity in this document series; and Brazil has, over the past two decades, overtaken the United States as the world's largest soybean producer and exporter, creating a genuine two-hemisphere supply structure in which the US and Brazilian harvest and export calendars alternate across the year. Understanding the US crop calendar, the crush and its two co-product markets, the China demand relationship, and the US-Brazil seasonal supply handoff is the foundation of spread and butterfly trading on the soybean curve.

Futures First — Internal Education Material
For Analyst Group Use

1. Contract Specifications and Delivery Mechanics

CBOT Soybeans is one of the most liquid agricultural futures contracts globally and the benchmark price reference for the soybean complex, which also includes the related Soybean Meal (ZM) and Soybean Oil (ZL) contracts described in Section 4.

Parameter	Detail
Exchange	CME Group — Chicago Board of Trade (CBOT)
Ticker	ZS (front month)
Contract size	5,000 bushels (approximately 136 metric tonnes)
Quotation	US cents per bushel (¢/bu); minimum tick = 1/4 cent/bu = $12.50 per contract
Contract months	January (F), March (H), May (K), July (N), August (Q), September (U), November (X) — seven delivery months per year, a notably denser calendar than corn or wheat's five months. November is the primary new-crop (harvest) contract.
Settlement type	Physical delivery via shipping certificate at CBOT-approved grain elevators and shipping stations in the Chicago/Illinois River delivery system, the same general delivery infrastructure described for corn.
Delivery grade	No. 2 Yellow soybeans at par, with other grades deliverable at exchange-set premiums and discounts, per USDA grading standards (test weight, damaged beans, foreign material, moisture, splits).
Trading hours	CME Globex: Sunday–Friday 19:00–07:45 ET and 08:30–13:20 ET, the same general electronic-session structure as corn and wheat.
Last trading day	Business day prior to the 15th calendar day of the delivery month
Contract value at $11.00/bu	$55,000 per contract
US marketing year	September 1 – August 31, the same convention as corn, reflecting the shared US autumn harvest timing.
Related contracts	Soybean Meal (ZM, CBOT, $/short ton) and Soybean Oil (ZL, CBOT, ¢/lb) — the two co-products of soybean processing, both described in Section 4. Together, ZS, ZM, and ZL constitute "the soybean complex."

1.1 Why Soybeans Have Seven Contract Months, Not Five

- Relative to corn and wheat's five-month annual cycle, soybeans list two additional contract months (January and August), reflecting both the contract's very high trading liquidity (supporting tighter calendar spacing across the curve) and the practical need for a contract month (August) that sits within the critical late-season US growing period, given soybeans' somewhat different — and, relative to corn, more drawn-out — critical-development-stage calendar described in Section 2.

1.2 Old-Crop / New-Crop Structure

- As with corn and wheat, soybeans divide into old-crop contracts (January, March, May, July — pricing the harvest already in storage) and new-crop contracts (August, September, November — pricing the crop not yet harvested or in progress), with November serving as the primary new-crop anchor in the same structural role December plays for corn. **The July-November spread** is the primary old-crop/new-crop transition spread for soybeans, directly analogous in function to corn's July-December spread.

2. The US Soybean Crop Calendar

US soybeans are planted and harvested on a single annual cycle broadly overlapping corn's calendar (the two crops are typically planted in close succession on the same farms, and often rotated year-to-year on the same fields), but soybeans' yield-critical period is somewhat later and more drawn-out than corn's sharply concentrated pollination window.

Period	Stage	Key Weather Risk	Primary Data Release	ZS Curve Impact
Mar–Apr	Planting intentions; early fieldwork preparation	Wet, cold soils delaying fieldwork (the same risk described for corn, given the shared planting window)	USDA Prospective Plantings (late March, covering both corn and soybeans)	First official acreage signal for new-crop November; directly relevant to the corn-soybean acreage competition described in Section 9.2
May–Jun	Planting (typically following corn planting by a short interval on most farms)	Persistent rain delaying planting; cold soils slowing emergence	USDA weekly Crop Progress (% planted vs 5-yr average); USDA June Acreage Report	Slow planting raises delayed-maturity and frost-exposure risk at the back end of the season; June Acreage Report is a major single-day event, as with corn
Jul	Vegetative growth and early flowering (R1–R2 stages)	Early-season drought stress; this period matters but is generally regarded as less acutely yield-decisive than the pod-set/seed-fill window the following month	USDA weekly Crop Progress (blooming %, condition ratings)	Building weather attention, though typically less acute than corn's simultaneous July pollination volatility
Aug	Pod set and seed fill (R3–R6 stages) — the single most yield-critical period	Drought and heat stress during pod set and seed fill directly reduces seed number and seed weight; this is soybeans' analogue to corn's July pollination window, but occurring roughly a month later and over a more extended multi-week period rather than a sharp single window	USDA weekly Crop Progress (pod-setting %, condition ratings); USDA August WASDE (first survey-based yield)	The most volatile weather-trading period of the year for ZS; the August WASDE is typically the single most anticipated scheduled report of the soybean crop year
Sep–Oct	Maturity and harvest	Early frost (can still reduce yield/quality if the crop has not fully matured); wet harvest conditions delaying fieldwork	USDA weekly Crop Progress (harvest %); monthly WASDE updates	Harvest pace confirms or revises the yield estimate; harvest-pressure selling, as with corn, typically weighs on cash basis and the front of the curve
Nov–Feb	Post-harvest marketing; on-farm and elevator storage	No growing-season weather risk domestically; Brazilian growing-season weather (Section 6) becomes the dominant live weather-risk variable globally during this window	USDA weekly export inspections and sales; monthly WASDE; Brazilian crop estimates and weather now the primary global supply-risk focus	Export pace vs USDA forecast; the market's attention progressively shifts toward Brazil as the US harvest is completed and marketed

2.1 Pod Set and Seed Fill — A More Drawn-Out Critical Window Than Corn's

- While corn's yield-critical pollination window is sharply concentrated into roughly two weeks, soybean yield formation during the R3 (pod set) through R6 (full seed) reproductive stages occurs over a more extended multi-week period in August, meaning soybean yield potential is sensitive to cumulative weather conditions across a longer window rather than to a single short, sharply-defined event in the way corn's pollination risk is concentrated — a structural difference that tends to make soybean weather-driven price volatility somewhat less sharply spiked, though still substantial, compared to corn's July pollination-window volatility.

3. The USDA Data Infrastructure

Soybeans are served by the same comprehensive USDA data infrastructure described in detail in the Corn reference document in this series — WASDE, NASS Crop Production surveys, weekly Crop Progress, and Quarterly Grain Stocks — since USDA collects and reports corn and soybean data through the same survey instruments and on a closely aligned publication calendar (the March Prospective Plantings, June Acreage, and September/January Grain Stocks reports, for example, cover both crops in the same publication). This section highlights soybean-specific features rather than repeating the full infrastructure description.

3.1 WASDE — The Crush-Inclusive Balance Sheet

- The soybean WASDE balance sheet includes a "crush" demand line (soybeans processed domestically into meal and oil) alongside exports and seed/residual use — a demand-side structure with no direct equivalent in the corn or wheat balance sheets, since neither of those crops has an analogous large-scale, two-co-product domestic processing demand category of comparable relative importance. The August, September, and October WASDE reports carry the same particular weight described for corn, as the first reports incorporating USDA's survey-based (rather than trend) yield estimate.

3.2 NOPA Monthly Crush Report

- The National Oilseed Processors Association (NOPA), an industry trade association representing the large majority of US soybean processing capacity, publishes a monthly crush report (soybeans processed, meal and oil production and stocks) roughly two weeks after each month-end — a higher-frequency, industry-sourced complement to the USDA's own monthly and WASDE-embedded crush estimates, and a closely watched report in its own right given its more granular and somewhat more timely publication relative to the official USDA crush figures.

4. The Crush — Meal, Oil, and Processing Economics

The "crush" — processing soybeans into soybean meal (the primary product by weight, used predominantly as a protein source in livestock and poultry feed) and soybean oil (used predominantly in food applications and, increasingly, biofuel) — is the central economic concept distinguishing the soybean market from corn and wheat, and the three-way price relationship between soybeans, meal, and oil is one of the most closely tracked processing-margin relationships in all of commodities trading.

4.1 The Crush Margin

- **What the crush margin represents**: a soybean processor's profitability is determined by the combined revenue from selling the meal and oil yielded by crushing a bushel of soybeans, less the cost of the soybean input itself (plus energy and other processing costs). Because one bushel of soybeans yields a roughly fixed proportion of meal and oil (approximately 44–48 lb of meal and 11 lb of oil from a 60 lb bushel, varying modestly with bean quality and processing technology), the crush margin can be calculated directly from the three CBOT contract prices (ZS, ZM, ZL) without needing any additional physical-market data — a calculation commonly referred to in the trade as the "board crush."
- **The board crush as a tracked and traded spread**: because the crush margin can be directly observed and replicated using exchange-traded contracts, it is both a real-world processor hedging reference (processors hedge their actual physical crush margin using offsetting ZS/ZM/ZL futures positions) and a directly tradeable spread relationship for speculative participants taking a view on processing economics independent of the flat-price direction of any one leg.

4.2 Meal Demand — Livestock Feed

- Soybean meal is the dominant protein component of US livestock and poultry feed rations (complementing corn as the dominant energy/caloric component, per the cross-reference in the Corn reference document), making meal demand directly tied to the same cattle, hog, and poultry sector dynamics described in the Corn reference document's feed-demand section.

4.3 Oil Demand — Food Use and the Growing Biofuel Pillar

- **Traditional food use**: soybean oil has historically been used predominantly in food applications — cooking oil, salad dressings, and as an ingredient in processed foods — competing with other vegetable oils (palm, canola/rapeseed, sunflower) in the broader global vegetable oil complex.
- **Renewable diesel and biodiesel — a rapidly growing demand pillar**: soybean oil is a primary feedstock for biodiesel and, increasingly, renewable diesel (a chemically distinct, higher-quality "drop-in" diesel substitute produced via a different refining process than traditional biodiesel), both incentivised by federal policy (the Renewable Fuel Standard's biomass-based diesel category, described in the same general RFS framework referenced for corn ethanol in the Corn reference document) and, in some cases, state-level low-carbon fuel standard programmes. The rapid build-out of US renewable diesel refining capacity over recent years has been a significant and closely tracked structural demand growth driver for soybean oil specifically.
- **The oil share of crush value**: the growing biofuel-driven demand for soybean oil has, at various points in recent years, increased oil's relative contribution to total crush revenue compared to its historical share, a structural shift tracked by analysts assessing the crush margin's longer-run drivers alongside the more traditional meal/feed-demand-driven framework.

5. Global Production — The US-Brazil-Argentina Triad

Unlike wheat's widely dispersed production geography, world soybean production is concentrated in a small number of countries, with the United States, Brazil, and Argentina together accounting for the substantial majority of global output and exports.

Country	Approx. Share of World Production	Harvest Period	Key Risk	ZS Curve Impact
Brazil	~38–40%, the largest single producer and exporter	Sep–Nov planting; Feb–May harvest (varies by region, Mato Grosso earliest)	Drought during pod-set/flowering (Dec–Feb); logistics congestion at export ports and along inland trucking routes during peak harvest/export season	The dominant Southern Hemisphere supplier; Brazilian harvest size and export pace are the primary driver of the global balance during the Northern Hemisphere winter/spring (Section 6)
United States	~28–32%	Apr–May planting; Aug pod-set/seed-fill; Sep–Oct harvest	Pod-set/seed-fill drought and heat stress (Section 2); spring planting delays	The dominant Northern Hemisphere supplier; sets the global price-discovery benchmark via CBOT during the Northern Hemisphere growing season and into early post-harvest marketing
Argentina	~12–14%	Oct–Dec planting; Mar–May harvest	Drought (a recurring and at times severe risk in major producing provinces); export tax and export quota policy (the same recurring Argentine policy variable described for corn and wheat)	A major producer and (especially) the world's leading exporter of soybean meal and oil, given Argentina's large domestic crushing industry — see Section 5.1
China	~4–5% production, but the dominant importer by far	Domestic harvest Sep–Oct, supplementing but far from meeting domestic crush demand	Domestic production weather is a minor global variable; Chinese import policy and demand pace are the dominant variable — see Section 7	China's role is overwhelmingly on the demand side of the global balance, not the supply side
Paraguay, other South American producers	~3–4% combined	Broadly aligned with the Brazilian/Argentine Southern Hemisphere calendar	Drought; logistics, given Paraguay's landlocked geography and reliance on river transport	A secondary but non-trivial South American supply contributor

5.1 Argentina's Distinctive Role — Crushing Rather Than Raw Bean Exports

- Argentina has historically differentiated its soybean export policy by applying higher export taxes on raw soybeans than on processed soybean meal and oil, an explicit policy choice intended to encourage domestic value-added processing rather than raw commodity export. The practical result is that Argentina has built a very large domestic crushing industry and is the world's leading exporter of soybean meal and soybean oil specifically, even though it exports a comparatively smaller share of its production as raw soybeans relative to Brazil and the US. This makes Argentine crush capacity utilisation, weather, and export tax policy a particularly important variable for the global meal and oil markets (ZM and ZL) specifically, somewhat distinct from its relevance to the raw soybean (ZS) balance.

6. Brazil — The Other Half of the Global Soybean Calendar

Because Brazil's soybean growing season falls almost exactly opposite the US calendar (Brazilian planting in Northern Hemisphere autumn, harvest in Northern Hemisphere winter/spring), the global soybean market effectively rotates through two roughly six-month "seasons of relevance" each year — US-centric from spring through autumn, Brazil-centric from autumn through spring.

6.1 The Brazilian Growing Season

- **Mato Grosso**: Brazil's largest soybean-producing state and typically the earliest to plant and harvest among major Brazilian growing regions, making Mato Grosso planting and harvest progress an early read on the broader Brazilian crop outlook.
- **Rio Grande do Sul**: a significant southern growing state with a somewhat different (more temperate, more variable) rainfall pattern than the tropical/savanna Mato Grosso region, and historically more exposed to drought risk in some El Nino/La Nina configurations than the central Brazilian growing belt.
- **The critical December-February flowering/pod-set window**: as with the US August window, Brazilian soybean yield is most sensitive to drought and heat stress during flowering and pod set, which for the Brazilian crop falls in the Southern Hemisphere summer, roughly December through February — the period when global market weather attention shifts decisively toward Brazil, just as the US crop is well past its own harvest.

6.2 CONAB and the Brazilian Data Infrastructure

- Brazil's National Supply Company (CONAB) publishes monthly soybean production estimates throughout the Brazilian growing and harvest season, the primary official Brazilian data source, tracked alongside private Brazilian agricultural consultancies and the USDA's own Brazilian production estimates embedded in the WASDE's world balance sheet.

6.3 Logistics — Ports and Inland Transport

- Brazilian soybean export logistics — moving the harvest from inland growing regions (particularly Mato Grosso, which is geographically distant from coastal ports) via truck, rail, and river barge to export terminals, predominantly at Santos and a growing number of northern Brazilian ports (Barcarena, Itaqui, and others, developed partly to reduce the trucking distance from Mato Grosso relative to the traditional Santos route) — have historically been a documented source of seasonal congestion and cost during the peak harvest/export window, making Brazilian port and logistics conditions a relevant basis and timing variable during the Brazilian export season.

7. China — The Dominant Demand Centre

China's role in the global soybean market is more concentrated and more singularly important on the demand side than any single country's role in almost any other commodity covered in this document series, making Chinese demand, import policy, and trade relations the single most important country-specific variable in the soybean fundamental picture.

7.1 Why China Dominates Global Import Demand

- China's soybean import volume — overwhelmingly used to feed its very large domestic crushing industry, which supplies both soybean meal for China's large hog and poultry sectors and soybean oil for Chinese food consumption — represents a substantial majority of total world soybean trade, a degree of demand concentration with few parallels among the commodities covered in this document series.
- **The Chinese hog sector connection**: because soybean meal's primary end use in China is hog and poultry feed, Chinese soybean import demand is structurally linked to the health and size of the Chinese hog herd specifically, including its periodic disruption by African Swine Fever (ASF) outbreaks — a disease risk affecting Chinese hog production that, when severe, can materially reduce Chinese soybean meal (and therefore raw soybean import) demand at the margin, a documented dynamic during the major 2018-2019 Chinese ASF outbreak period.

7.2 China's Sourcing Split Between the US and Brazil

- China sources soybeans from both the US and Brazil (and, to a smaller extent, Argentina), and the seasonal pattern of Chinese purchasing — weighted toward US-origin supply during the US export season (autumn through winter) and toward Brazilian-origin supply during the Brazilian export season (spring through summer) — mirrors and reinforces the two-hemisphere seasonal calendar described in Section 6, with China's purchasing pattern itself shifting between origins as each hemisphere's new crop becomes available.
- **US-China trade relations as a direct price variable**: US soybean exports to China have, at various points, been directly affected by US-China trade policy actions, including tariffs and retaliatory tariffs imposed during the 2018-2019 trade dispute period, which significantly reduced US soybean exports to China for a period and accelerated Chinese sourcing diversification toward Brazilian supply — a documented historical instance of trade policy directly and substantially reshaping established global soybean trade flows, analogous in mechanism to the trade-policy sensitivity described for US cotton exports to China elsewhere in this document series.

7.3 Chinese State Reserve and Import Administration

- Chinese state entities manage strategic soybean reserves and administer import licensing and quota mechanisms (including, historically, requirements related to genetically modified organism approval for imported soybean varieties), creating a policy layer that can affect the pace and sourcing of Chinese import demand independent of pure price or weather fundamentals.

8. The Forward Curve Structure — Zones and Spread Mechanics

The soybean forward curve reflects the same general cost-of-carry framework as corn, modulated by the US crop calendar (Section 2) and — distinctively — by the Brazilian growing and harvest season (Section 6), which provides a partially offsetting global supply pulse roughly six months out of phase with the US crop, in a structural role broadly analogous to (though larger in absolute global market-share terms than) the Brazilian safrinha corn relationship described in the Corn reference document.

Contract	Crop-Year Role	Primary Driver	Secondary Driver	Key Signal
November (X)	New-crop anchor. The primary contract referencing the US harvest just completed.	US planted acreage; growing-season weather (especially August pod-set/seed-fill); August WASDE first survey-based yield	Early Brazilian planting-season outlook (a partially offsetting global supply signal arriving roughly four months later)	USDA Prospective Plantings; June Acreage; weekly Crop Progress through pod-set; August/September/October WASDE
January (F)	New/old-crop bridge. Post-harvest US marketing; Brazilian planting/early growth underway.	US export sales and shipment pace vs USDA forecast; early Brazilian crop condition reports (planting now complete, early vegetative growth)	Chinese purchasing pace, now beginning to weigh US versus upcoming Brazilian-origin sourcing	Weekly export sales/inspections; CONAB early-season Brazilian estimates
March (H)	Old-crop. Brazilian crop now in its critical Dec-Feb flowering/pod-set window, results becoming clearer.	Brazilian weather during and immediately after its critical pod-set window (Section 6.1); early Brazilian harvest beginning in the earliest-maturing regions (Mato Grosso)	US old-crop export pace and ending-stocks trajectory	CONAB monthly estimates; Brazilian weather forecasts; early Mato Grosso harvest yield reports
May (K)	Old-crop, deep. Brazilian harvest in full swing and largely confirming that season's production.	Brazilian harvest pace, yield confirmation, and export logistics/port congestion (Section 6.3)	US new-crop planting intentions firming ahead of the March/April acreage data	CONAB mid/late-harvest estimates; Brazilian port and logistics reports; Chinese purchasing shift toward Brazilian-origin supply
July (N)	Last old-crop contract. US new crop entering its own July vegetative /early-flowering stage.	Old-crop/new-crop (July-November) spread dynamics (Section 1.2); US planting-to-date conditions setting up the new-crop outlook	Confirmed Brazilian export pace for the season now largely known	Weekly US Crop Progress; old-crop ending-stocks confirmation via June/September Grain Stocks
August (Q)	New-crop, early. US crop entering its critical pod-set/seed-fill window concurrently with this contract.	US pod-set/seed-fill weather (the single most volatile input of the year, per Section 2.1)	August WASDE (first survey-based yield, released mid-month)	Daily/weekly US temperature and precipitation forecasts during pod-set; August WASDE

8.1 Carry, Basis, and the China-Driven Export Basis Pattern

- As with corn, harvest-pressure cash basis weakness during the September-November US harvest window is a well-documented seasonal pattern, gradually firming as the crop is absorbed into storage and export channels. Export basis at US Gulf and Pacific Northwest terminals is additionally sensitive to the pace and concentration of Chinese purchasing specifically, given China's outsized share of total demand described in Section 7 — a notably more concentrated single-buyer influence on export basis than exists for corn or wheat, whose export demand is spread across a more diversified set of importing countries.

9. Macro and Cross-Market Drivers

9.1 The Soybean Oil Share and Crude Oil/Biofuel Linkage

- As described in Section 4.3, the growing renewable diesel and biodiesel demand pillar connects soybean oil (and, through the crush relationship, soybeans themselves) to crude oil and diesel pricing and to biofuel policy, in a mechanism structurally analogous to — though operating through a different feedstock and policy sub-category than — the corn ethanol/RFS linkage described in the Corn reference document.

9.2 The Corn-Soybean Price Ratio and Acreage Competition

- As described from the corn side in the Corn reference document, corn and soybeans compete directly for the same US farmland, and the corn-soybean price ratio at planting-decision time influences the acreage split reflected in the March Prospective Plantings and June Acreage reports — the identical mechanism viewed from the soybean side of the relationship.

9.3 The Global Vegetable Oil Complex

- Soybean oil competes with, and its price is influenced by, the broader global vegetable oil complex — particularly palm oil (the world's most-consumed vegetable oil by volume, produced predominantly in Indonesia and Malaysia) and canola/rapeseed oil (produced predominantly in Canada and the EU) — since food manufacturers and, increasingly, biofuel producers can substitute among vegetable oils to some degree based on relative price, making palm oil price trends and Indonesian/Malaysian palm oil policy (including export tax and biodiesel mandate policy in those countries) a relevant cross-market input to the soybean oil (and, through the crush, soybean) price.

9.4 US Dollar and Export Competitiveness

- As a globally traded, USD-denominated commodity with substantial export demand, soybeans exhibit the standard dollar-priced-commodity relationship described elsewhere in this document series, with the added dimension that BRL/USD and ARS/USD (Brazilian Real and Argentine Peso) movement directly affects the relative export competitiveness of the two largest Southern Hemisphere competing origins against US supply.

9.5 Speculative Positioning — COT Data

- **CFTC COT for CBOT Soybeans**: weekly managed money net positioning is closely tracked, following the same general crowding/reversal-risk framework described throughout this document series. Soybean positioning is also tracked alongside the related soybean meal and oil COT data given the crush-margin relationship described in Section 4.1, with some participants monitoring the relative positioning across all three legs of the crush as a more complete picture of speculative sentiment toward the complex.

10. Documented Seasonal Patterns

Soybeans' seasonal pattern is shaped by the alternation between US-centric and Brazil-centric attention windows described in Section 6, layered onto the same general USDA-report-driven calendar structure described for corn.

Period	Dominant Geography	Key Driver	Typical ZS Behaviour	Reliability
Jan–Feb	US post-harvest marketing transitioning to Brazil-focus	US export pace; early Brazilian crop condition as the Dec-Feb critical window concludes	Attention shifting decisively toward Brazil; early signs of Brazilian crop stress or favourable conditions can move the market significantly	Moderate
Mar–May	Brazil-dominant (harvest)	Brazilian harvest pace, yield confirmation, port logistics/congestion	CONAB and private Brazilian estimates are the primary catalysts; a confirmed large or small Brazilian crop can set the tone for the rest of the global marketing year	Moderate–High
Jun	US-dominant (planting)	US planting progress; June Acreage Report	A major single-day event around the Acreage Report, as with corn	Moderate, spiking on report day
Aug	US-dominant (pod-set/seed-fill)	US weather during the critical reproductive window; August WASDE	The most volatile weather-trading period of the year for ZS	High — the window itself is calendar-certain even though any specific weather outcome is not
Sep–Nov	US-dominant (harvest)	Harvest-pressure basis weakness; September WASDE/Grain Stocks data day; yield confirmation	Similar harvest-pressure pattern to corn, moderated by on-farm storage capacity	Moderate–High
Dec	Transition toward Brazil-focus	Early Brazilian planting/early-season weather; year-end fund positioning and index-roll mechanics	The handoff period as US harvest marketing winds down and Brazilian-season attention begins building	Moderate

10.1 Documented Historical Events

- **2012 US drought**: the same severe US drought described in the Corn reference document also significantly reduced the 2012 US soybean crop, given the overlapping geography and timing of the two crops' growing seasons, contributing to record nominal soybean futures prices alongside corn's comparable price spike that year.
- **2018-2019 US-China trade dispute**: as detailed in Section 7.2, US-China tariff actions during this period substantially reduced US soybean exports to China for a period, depressed US soybean prices relative to where they would otherwise have traded given the supply/demand fundamentals, and accelerated a structural shift in Chinese sourcing toward Brazilian supply that persisted beyond the resolution of the immediate trade dispute.
- **2018-2019 Chinese African Swine Fever outbreak**: as referenced in Section 7.1, a severe ASF outbreak that significantly reduced the Chinese hog herd over this period reduced Chinese soybean meal demand at the margin, compounding the trade-dispute-driven reduction in Chinese soybean import demand occurring concurrently.

- **Periodic Argentine drought episodes**: Argentina's soybean crop has, at various points, experienced severe drought (including a notably severe and well-documented episode affecting the 2022/23 Argentine crop), significantly reducing Argentine production and export availability and tightening the global soybean meal and oil balance specifically, given Argentina's outsized role in those two markets described in Section 5.1.

11. Key Data Sources and Release Schedule

Report	Publisher	Frequency	Typical Timing	Primary ZS Curve Impact
WASDE (Soybean component)	USDA WAOB	Monthly	~11th–12th of month, 12:00 ET	All contracts — monthly balance-sheet revision including the crush demand line; August–October editions carry particular weight
USDA Prospective Plantings	USDA NASS	Annual	Late March	November — first acreage signal, directly relevant to corn-soybean acreage competition (Section 9.2)
USDA Acreage Report (June)	USDA NASS	Annual	Last business day of June	November — confirms/revises planted acreage; a major single-day mover, as with corn
USDA Quarterly Grain Stocks	USDA NASS	Quarterly	Late Mar, Jun, Sep, Jan	Front 1–2 contracts — independent stocks measurement, can reveal demand-estimate errors
USDA Weekly Crop Progress	USDA NASS	Weekly (Apr–Nov)	Monday ~16:00 ET	Front 1–2 contracts during the growing season — condition ratings through the August pod-set window most watched
USDA Weekly Export Sales / Inspections	USDA FAS / AMS	Weekly	Thursday 08:30 ET (sales); Monday (inspections)	C1–C3 — export pace vs WASDE forecast; China-specific sales are by far the most closely watched single data line
NOPA Monthly Crush Report	National Oilseed Processors Association	Monthly	~3rd–5th business day of month	C1–C3 — higher-frequency, industry-sourced crush data complementing USDA's figures
CONAB Brazilian crop estimates (soybean component)	CONAB (Brazil)	Monthly	~3rd week of month	C3–C8 — the primary official Brazilian production data source during the Southern Hemisphere season
Argentine Bolsa de Cereales crop estimates	Bolsa de Cereales de Buenos Aires	Weekly/monthly	Variable	C3–C8 — Argentine crop and weather updates, particularly relevant to the meal/oil (ZM/ZL) balance
Chinese customs import data	China General Administration of Customs	Monthly	Mid-month	All contracts — confirms actual Chinese import volume against trade and USDA forecasts
NOAA/CPC outlooks (US and South American relevant regions)	NOAA CPC	Daily/continuous	Continuous	Front contract, acutely during US Aug pod-set and Brazilian Dec-Feb flowering windows
CFTC Commitments of Traders (Soybeans, Meal, Oil)	CFTC	Weekly	Fri ~15:30 ET (3-day lag)	Positioning / crowding — tracked across all three legs of the crush given the margin relationship

CBOT Soybeans occupy a structurally distinct position among the major row crops covered in this document series, defined by three features with no exact parallel in corn or wheat: the crush's three-way processing-margin relationship between soybeans, meal, and oil; China's uniquely concentrated role as the dominant single demand centre, making Chinese policy and demand pace a more singularly important country-specific variable than for almost any other commodity in this series; and a genuine, large-scale two-hemisphere production structure in which Brazil — now the world's largest producer — provides a roughly six-months-out-of-phase supply pulse that shifts the market's fundamental attention between the US and South America over the course of each year.

For spread traders, the most reliable recurring framework is to track which hemisphere's growing season currently has the market's attention (US-dominant roughly April through November, Brazil-dominant roughly December through May); to monitor the July-November old-crop/new-crop spread as the US-specific structural read; to track the board crush margin as both a processor-hedging reference and an independently tradeable relationship; and to weight Chinese demand-pace and trade-policy developments more heavily than the single-country demand weight typically warranted in most other commodity markets covered in this series, given China's outsized concentration on the demand side of the global soybean balance.